

OneGemmy

Entity Relationship Diagram & Multi-Tenancy Architecture

FastAPI + SQLAlchemy + SQLite | Feature-Grouped API v1

13 Tables | 48 Permissions | 7 Roles | 5 Demo Users

Gemmy Connect Ltd

Built with care in Rwanda

Tenant Hierarchy

Platform	[super_admin]	Manages all companies, users. Global access. No company_id filter.
Company	[company_admin]	Tenant boundary. Owns shops, roles, employees. Scoped to company_id.
Shop	[user]	Operational unit within company. Owns products, transactions. Scoped to shop_id.
User	[—]	Belongs to a company + shop + role. Role determines permission set.
Role	[—]	Per-company or system role. Links to 48 granular permissions via M2M.
Permission	[—]	Atomic access unit: module.action (e.g. stock.view, sales.create).

Data Isolation Strategy

- All business tables (Product, Deal, Lead, Contact, Transaction, Project, Employee) are scoped via company_id.
- Users see only their company's data. Super admins bypass all filters.
- Roles are per-company (company_id = NULL for system roles).
- Shops are per-company. A user can be assigned to one or more shops.
- get_current_user() loads user + role + permissions on every request.
- require_permission() checks the loaded permissions list. Returns 403 if missing.

Request Flow (RBAC)

1. Client sends JWT token in Authorization header.
2. get_current_user() decodes token → loads User from DB.
3. require_permission("stock.view") checks: user.role.permissions contains "stock.view"?
4. If yes → route handler runs. If no → 403 Forbidden.
5. Super admins bypass all permission checks (platform_role == "super_admin").
6. Company admins get Owner role permissions (all 48 perms).

Seed Data (Demo)

Roles: Owner (48), Admin (40), Manager (30), Sales Rep (10), Inventory Mgr (12), Accountant (14), Employee (8)

Users: super@onegemmy.com (super_admin), admin@onegemmy.com (Owner), manager@onegemmy.com (Manager), sales@onegemmy.com (Sales Rep), staff@onegemmy.com (Employee)

ERD — Auth & Tenancy Tables

companies		
id	INTEGER	PK
name	VARCHAR(255)	
slug	VARCHAR(255)	UNIQUE
plan	VARCHAR(50)	DEFAULT 'free'
is_active	BOOLEAN	DEFAULT true
created_at	TIMESTAMP	SERVER DEFAULT

shops		
id	INTEGER	PK
company_id	INTEGER	FK → companies.id
name	VARCHAR(255)	
location	VARCHAR(255)	
status	VARCHAR(50)	DEFAULT 'active'
created_at	TIMESTAMP	SERVER DEFAULT

roles		
id	INTEGER	PK
company_id	INTEGER	FK → companies.id
name	VARCHAR(255)	1 : N
description	TEXT	
color	VARCHAR(20)	DEFAULT '#64748B'
is_system	BOOLEAN	DEFAULT false
created_at	TIMESTAMP	SERVER DEFAULT

permissions		
id	VARCHAR(100)	PK
name	VARCHAR(255)	
description	TEXT	
module	VARCHAR(100)	

users		
id	INTEGER	PK
email	VARCHAR(255)	UNIQUE
name	VARCHAR(255)	
hashed_password	VARCHAR(255)	
platform_role	VARCHAR(50)	DEFAULT 'user'
created_at	TIMESTAMP	SERVER DEFAULT

Legend:  PK  FK  Crow's Foot (Many)  Relationship

products		
id	INTEGER	PK
sku	VARCHAR(100)	UNIQUE
name	VARCHAR(255)	
description	TEXT	
category	VARCHAR(100)	
price	FLOAT	
cost	FLOAT	DEFAULT 0
stock_quantity	INTEGER	DEFAULT 0
min_stock	INTEGER	DEFAULT 0
is_active	BOOLEAN	DEFAULT true
created_at	TIMESTAMP	SERVER DEFAULT
updated_at	TIMESTAMP	ON UPDATE

leads		
id	INTEGER	PK
name	VARCHAR(255)	
email	VARCHAR(255)	
phone	VARCHAR(50)	
company	VARCHAR(255)	
source	VARCHAR(100)	
notes	TEXT	
is_converted	BOOLEAN	DEFAULT false
created_at	TIMESTAMP	SERVER DEFAULT
updated_at	TIMESTAMP	ON UPDATE

deals		
id	INTEGER	PK
title	VARCHAR(255)	
value	FLOAT	
stage	VARCHAR(50)	DEFAULT 'lead'
probability	INTEGER	DEFAULT 0
lead_id	INTEGER	FK → leads.id
owner_id	INTEGER	FK → users.id
notes	TEXT	
closed_at	TIMESTAMP	
created_at	TIMESTAMP	SERVER DEFAULT
updated_at	TIMESTAMP	ON UPDATE

transactions		
id	INTEGER	PK
description	VARCHAR(255)	
amount	FLOAT	
type	VARCHAR(50)	
category	VARCHAR(100)	
reference	VARCHAR(100)	
created_at	TIMESTAMP	SERVER DEFAULT

Legend: — PK ■ FK ● Crow's Foot (Many)

1 : N

- - - Relationship

projects		
id	INTEGER	PK

employees		
id	INTEGER	PK

Roles & Permissions

Role	Perms	Modules	Description
Owner	48	All	Full access to everything. Company creator.
Admin	40	All except user mgmt	Manages everything except inviting users.
Manager	30	Sales, Stock, Finance, Projects	Day-to-day operations across modules.
Sales Rep	10	CRM, Sales (view/create/edit)	Manages leads, deals, contacts.
Inventory Mgr	12	Stock (full), Products	Manages products, stock levels.
Accountant	14	Finance (full), Reports	Manages transactions, invoices, reports.
Employee	8	Projects (view), Contacts (view)	Limited read access. No sensitive data.

Permission Modules (48 total)

dashboard	view
sales	view, create, edit, delete, leads.manage, quotes.manage
stock	view, create, edit, delete
finance	view, create, edit, delete
projects	view, create, edit, delete
crm	view, contacts.manage, interactions.log
hr	view, employees.manage
reports	view
settings	view, users.manage, roles.manage